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Highly scalable machine vision enabled with meta-optics-based ultra-wide neural network

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Abstract

Optical neural networks (ONNs) offer a route to low-latency, energy-efficient AI, but scaling them to modern model sizes is constrained by two practical bottlenecks: training large ONNs is computationally prohibitive, and implementing or tuning millions of optical components is highly sensitive to fabrication imperfections and alignment errors. Here we report a metasurface-based optical learning machine that bypasses these barriers by operating in an ultra-wide regime. We use an optical metasurface with 41 million optical parameters to form an ultra-wide NN, and show that, at this scale, a fixed, untrained metasurface can closely approximate a fully trained one. Residual mismatch is compensated by a compact digital backend with only 10^2 – 10^4 trainable parameters, enabling task adaptation without retraining or re-fabricating the metasurface. Experimentally, the resulting hybrid system delivers scalable machine vision across six tasks, achieving accuracy competitive with state-of-the-art models including ResNet and Vision Transformer, while remaining tolerant to fabrication imperfections and misalignment. Our approach is highly scalable, tolerant to fabrication errors, and operates under both coherent and incoherent illumination, thus providing a practical pathway to large-scale, high-performance optical AI computing.

Keywords Optical neural network, Neuromorphic computing, Machine vision

1 Introduction

Neural networks (NNs) have become a powerful tool across various scientific and technological domains, triggering transformative shifts in fields such as drug discovery, image processing, autonomous vehicles, and medical diagnostics [1]. However, the growing complexity

of problems is causing the cost of training and inferring NN models to double every 2 months [2], significantly outpacing the advancements in complementary metal-oxide-semiconductor (CMOS) circuits. Optical neural networks (ONNs) are highly promising for addressing the above challenges due to the high degree of parallelism inherent in light waves [3–46]. By harnessing this parallelism, ONNs can offer significant energy efficiency and speed advantages.

Despite the theoretical potential of ONNs, current implementations face challenges in scaling up parameter numbers. Two-dimensional (2D) integrated ONNs can achieve high computing speed but have a limited number of components, due to the large footprint of optical components, the insertion loss, and the complexities involved in controlling these components. By transitioning to 3D free-space optics, spatial parallelism in optics allows for

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a significant increase in the number of parameters [5]. Experimental demonstrations have successfully realized approximately 100,000 scalar multiplications [39–46]. However, the parameter count is still significantly smaller than current AI models, which may have millions to billions of parameters [47, 48].

An appealing solution to further scale up ONNs is utilizing optical metasurfaces, which consist of high-density meta-atoms arranged in a two-dimensionally periodic structure [49–55]. Each meta-atom can be precisely tailored to modulate the transmission property of incident light at the sub-wavelength scale. With a density of more than 4 million meta-atoms per mm^2 , optical metasurfaces have the potential to enable a large-scale ONN with the parameter number close to that in large-scale AI models. However, current metasurface-based ONN demonstrations still have very limited parameter numbers, fewer than 1 million [37, 38, 42, 43, 56]. Training large-scale ONNs via gradient descent is computationally prohibitive because accurate gradient computation requires complex physics-based models, leading to long training times, high memory costs, and large optimization spaces. Even when such training is achieved, slight misalignments at sub-wavelength scales can result in significant reductions in accuracy [57, 58]. Moreover, because metasurfaces are typically not programmable, every new task requires re-training and fabricating a new metasurface chip, which limits their deployment in practical applications.

Alternative computing paradigms can eliminate the need for large-scale training. Examples include reservoir computing [59–75], extreme learning machines [76–80], and random projections [68, 81–83]. These computing systems combine a fixed optical system as the optical frontend, together with a trainable digital layer as the backend. By only training the digital layer, the overall computing system is adaptable to various downstream tasks. However, current demonstrations mostly rely on the intrinsic physics of optical systems. The performance of the optical frontend cannot be guaranteed. As a result, the overall performance often falls short of fully trained neural networks. To compensate for the non-optimal performance of the optical frontend, an excessively large digital neural network is required, undermining the energy efficiency and speed advantages that optics aim to provide.

To address these limitations, this work employs a large-scale optical metasurface composed of 41 million meta-atoms to implement an optical frontend, inspired by the extreme learning machine (ELM) [84–86]. In a classic ELM, the network consists of a single hidden layer whose parameters are randomly initialized and kept fixed, while only the output layer is trained. Subsequent variants extend this framework by modifying

the network depth and architectural structure. The performance of an ELM depends on the width of the hidden layer: increasing the number of hidden neurons enhances the representation capacity and improves accuracy [35, 87–90]. Therefore, we exploit an optical metasurface with 41 million meta-atoms to realize an ultra-wide hidden layer, which is then followed by an ultra-compact digital NN serving as the output layer. Together, these components form a hybrid ELM architecture that leverages massive optical parallelism in metasurface for high-capacity feature extraction while retaining digital flexibility for task-specific learning. Further, guided by the Neural Tangent Kernel (NTK) framework [91], which provides a theory of dynamics in wide NN, we initialize the weights using a Gaussian distribution with an appropriately chosen variance (the variance selection is discussed in Supplementary Note 2). In this regime, NTK theory predicts that training induces only minimal changes to the hidden-layer parameters; consequently, fixing the optical frontend and training only a compact digital NN can enable close to optimal task-specific learning compared to a fully trained system. Consistent with this prediction, our hybrid system, which uses a digital NN with only 10^2 to 10^4 trainable parameters, successfully addresses six diverse tasks and achieves accuracy that surpasses prior ONNs and is competitive with leading vision models such as ResNet and Vision Transformer. This includes real-world applications such as large-scale breast cancer diagnosis and lesion localization. A detailed comparison of our system with other prior ONNs is provided in Table S9 in Supplementary Note 27.

From a practical implementation perspective, our metasurface-based hybrid system eliminates the need for precise training individual meta-atoms. Moreover, the system is highly tolerant to fabrication imperfections and alignment errors, as their effects can be readily compensated by retraining the compact digital NN. These features confer distinct scalability advantages: (1) the system scales naturally to high-resolution inputs, as demonstrated by end-to-end processing of medical images with more than 10^6 pixels; (2) it readily extends to more complex NN architectures: for example, we implement a recurrent NN (RNN) that uses the hybrid system as its internal state, achieving 99.1% accuracy on an action recognition task. Furthermore, our system can operate under both coherent and incoherent illumination, broadening the application scope beyond conventional ONNs that rely on coherent light. With over 99.99% of the computation performed passively by the optical metasurface (see Supplementary Note 4 for the derivation), the system achieves exceptional energy

efficiency and low latency. Overall, our system offers a highly practical pathway toward high-performance optical AI computing.

2 Results

2.1 Working principle and device design

The implementation of the proposed metasurface-based optical learning machine (MOLM) is illustrated in Fig. 1a. The MOLM consists of a metasurface-based optical frontend, followed by a compact digital NN that can be trained to generate prediction results for various tasks. The optical frontend is an optical metasurface created by $6,400 \times 6,400 = 41$ million silicon cylindrical meta-atoms. Their diameters range from 100 to 400 nm, with a unit cell period of 500 nm. The diameter of each meta-atom is determined in two steps. First, we generate a $6,400 \times 6,400$ matrix of optical phase modulation coefficients, randomly sampled from a Gaussian distribution with zero mean and a standard deviation (SD) of 0.4π . The diameter of each meta-atom is then obtained from the mapping between the optical phase modulation coefficient and meta-atom diameter, as detailed in Supplementary Note 1. The details of metasurface design and fabrication are provided in Methods and Supplementary Notes 1 and 3. An optical lens is placed between the metasurface and a CMOS sensor to perform a Fourier transform, which further enhances the network's ability to extract high-frequency features. The CMOS sensor, with a resolution of 600×800 , is placed at the focal plane of the optical lens to bridge the optical frontend and the digital backend through photodetection while providing square nonlinearity. A 532-nm laser diode is used as the light source for the following experiments. As shown in the Discussion, our system can also operate under incoherent LED illumination. The sensor output is downsampled before being fed into the digital NN to ensure compactness, and the network is trained on these downsampled outputs to generate final predictions.

Before photodetection, the input optical image $\mathbf{U}_0$ undergoes a linear transformation given by $\mathcal{F}(\mathbf{H}_d \mathbf{U}_0 \otimes \mathbf{W})$, where $\mathbf{W}$ is the complex-valued transmission matrix of the metasurface, $\mathbf{H}_d$ is the diffraction matrix, $\mathcal{F}(\cdot)$ denotes the Fourier transform implemented by the optical lens, and $\otimes$ represents the Hadamard product. Diffraction induces an effective $N \times N$ fully connected network, where N is the number of meta-atoms. After photodetection, the output of the k -th sensor pixel is given by $\tilde{U}_k = \left[\sum_{i=k \cdot (R-1)}^{k \cdot R} \mathcal{F}(\mathbf{H}_d \mathbf{U}_0 \otimes \mathbf{W})[i, :] \right]^2$, where R is the ratio of the number of meta-atoms to sensor pixels ($R = N/M$). The photodetector implements a square-law

operation, which serves as the nonlinear activation (neuron) in the hidden layer of an ELM. The followed digital NN serves as the output layer of the ELM. Therefore, our system is equivalent to an ELM in architecture. See the deviation details of the system model in Supplementary Note 5.

To illustrate the importance of the parameter N , we visualize the transformed images as N varies, as shown in Fig. 1c. Information entropy has been used to evaluate the performance of NNs, including ELMs [35, 92, 93]. Increasing N significantly enhances the information entropy, from 8.4 bits without the metasurface to 13 bits with 41 million meta-atoms, quantified as $-\sum_{k=0}^M \bar{U}_k \log_2(\bar{U}_k)$, where $\bar{U}_k = \tilde{U}_k / \sum_{k=0}^M \tilde{U}_k$ [94], as shown in Fig. 1d. In addition, Fig. 1e shows that the transformed images exhibit increasingly rich high-frequency components, which, according to machine learning theory [95], enable networks to learn high-frequency functions in low-dimensional domains. A detailed explanation of the enhancement of high-frequency components and information entropy is provided in Supplementary Note 26. This enhanced ability to capture high-frequency features improves MNIST classification accuracy from 87.3% without the metasurface to 94.7% with 41 million meta-atoms, even in the absence of an optical lens, as shown in Fig. 1g. When an optical lens is introduced to perform a Fourier transform, these high-frequency features are further amplified, leading to an accuracy improvement from 94.7 to 99.2%. These results are obtained by downsampling the output of a 600×800 sensor array to 15×20 , enabling a compact digital NN with only 300×10 weights. We also validate our approach on the more complex CIFAR-10 dataset, observing a similar trend: accuracy increases as the number of optical parameters grows to 10^7 . The final experimental accuracy reaches 91.6% with only 7,440 trainable digital parameters, as shown in Fig. 1h, which is comparable to that of ResNet-50. The experimental details about MNSIT and CIFAR-10 benchmarking can be found in Methods and Supplementary Note 16.

We further show that when the metasurface contains tens of millions of parameters in the hidden layer, a fixed, untrained metasurface approaches the performance of a fully trained one. This is validated by comparing the root mean square error (RMSE) between the parameters in untrained and trained networks with the same model architecture. Fig. 1f shows that the RMSE between the untrained and trained optical parameters decreases as the number of optical parameters N increases. When N is 41 million, the RMSE reduces from 1.02 to 0.018. This scaling behavior is consistent with predictions from NTK theory. The residual difference between the untrained and trained optical front-ends can be effectively

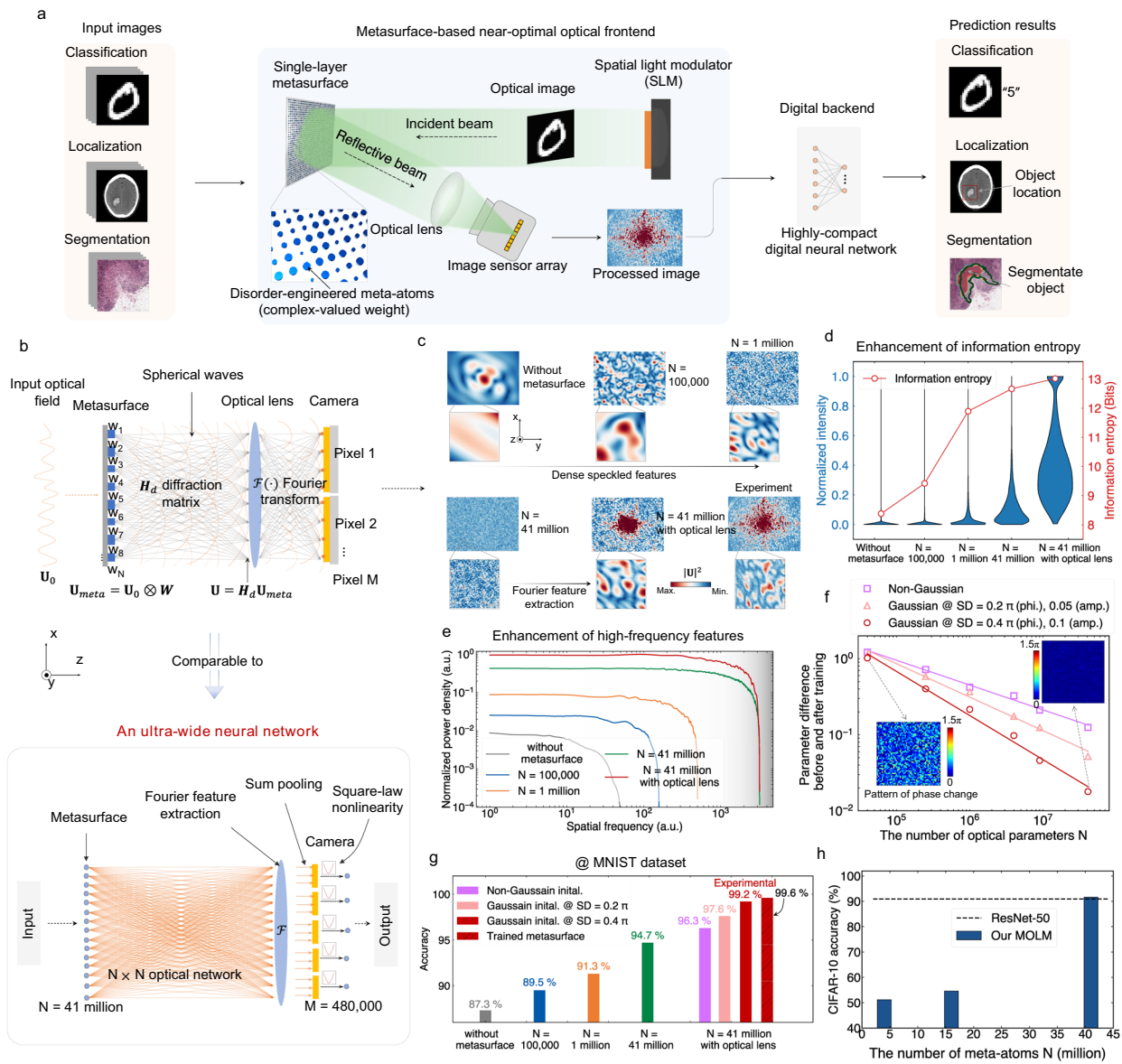


Fig. 1 The metasurface-based optical learning machine (MOLM). **a** Schematic of the MOLM. The optical image generated by the SLM is projected onto a single-layer metasurface consisting of massive cylindrical silicon meta-atoms. An optical lens then collects the reflected optical image by the metasurface. Following that, the optical field at the focusing plane is captured by an image sensor array. Lastly, the captured image is fed into a digital neural network to produce the prediction result. **b** The illustration of the metasurface-based ultra-wide neural network. **c** The transformed images by the metasurface-based optical frontend versus the number of meta-atoms N . **d** The value's distribution and information entropy of transformed images versus the number of meta-atoms N . **e** The normalized power density of the transformed image versus the spatial frequency for different numbers of meta-atoms N . **f** The parameter difference (RMSE) before and after training versus the number of optical parameters N . **g** The MNIST accuracies of the metasurface-based optical frontend with different configurations. **h** The experimental CIFAR-10 accuracy versus the number of meta-atoms

compensated by a small digital NN. As a result, the overall MNIST classification accuracy achieved by the untrained network reaches 99.2%, which is very close to the fully trained accuracy of 99.6%, as shown in

Fig. 1g. Such scaling behavior is also influenced by the initialization statistics. When the optical parameters are drawn from a narrower Gaussian distribution ($SD = 0.2\pi$) or from a non-Gaussian distribution extracted

from the mode coupling of multimode fibers [76], the RMSE between untrained and trained optical parameters still decreases with increasing parameter count, but at a slower rate. This suggests that the initialization statistics are tunable hyperparameters to optimize, but they are typically empirical and depend on the specific system and task. The choice of $SD = 0.4\pi$ in our design is further justified in the Supplementary Note 2.

2.2 Experimental demonstration of high performance, versatility, and practicability

We begin by conducting machine vision tasks to demonstrate the high performance, versatility, and high scalability of our system. For benchmarking, we compared the performance of our MOLM with three large-scale deep learning models: ResNet-50 [96], a classical 50-layered CNN with approximately 23.5 million parameters; the Segment Anything model (SAM) [97], a cutting-edge large promotable segmentation model with 93.7 million parameters; and Vision Transformer (ViT) [47], a transformer encoder model for image classification with more than 85.8 million parameters (Supplementary Note 18). Figure 2a shows the schematic of the experimental setup for benchmarking characterization (see details in Methods and Supplementary Note 4). For each task, the input images are generated using an SLM. These images are processed by the metasurface and then collected by an optical lens before being detected by a CMOS digital camera. The detected digital image is downsampled, with the downsampling ratios being task-dependent. The same metasurface chip is used for all the tasks. Only the digital neural network at the backend is trained for different applications.

The tasks are chosen not only to demonstrate our system's ability to adapt to diverse applications using a fixed, untrained metasurface, but also to highlight its practical advantages in implementation. Specifically, we showcase its capabilities in: (1) processing with

an ultra-compact digital neural network (COVID-19 Radiography), (2) handling high-resolution images in a single shot (Thoracic Disease Detection), and (3) performing multitask classification and detection by redesigning only a small digital NN.

2.2.1 Image classification with ultra-compact digital NN

COVID-19 Radiography: We first demonstrate the application of our MOLM on the COVID-19 Radiography dataset [98], as shown in Fig. 2e, highlighting its ability to achieve high-performance image classification using a highly compact digital neural network. This dataset includes over 20,000 chest X-ray (CXR) images covering normal and COVID-19 positive cases, each with 299×299 pixels. The optical images with 299×299 pixels are generated by the SLM, processed by the MOLM, and subsequently detected. The detected image is dramatically downsampled to only 12×16 pixels, followed by a highly compact regression network with only 192 weights at the digital backend to produce the binary classification results for normal and COVID-19 cases. The accuracy, precision, sensitivity, and specificity obtained from the classification results are 98.0%, 96.8%, 99.2%, and 96.9% using only 192 digitally trained weights, respectively. The experimentally obtained accuracy of MOLM up to 98.0%, outperforms previously demonstrated simulation and experimental ONNs [76, 99], as shown in Fig. 2g. This accuracy is competitive against ResNet-50 (97.0%) and SAM (97.2%) and slightly lower than ViT (99.0%). Nevertheless, our MOLM can compress the digital model by a factor of 4.9×10^5 compared to ResNet-50 and 1.8×10^6 compared to ViT, signifying a remarkable reduction of computing time by 1.2×10^5 times at the training stage and energy consumption by 1,900 times at the inference stage.

The computing time includes the time required for SLM response, the free-space propagation of the light, camera response, and executing a digital computer,

(See figure on next page.)

Fig. 2 Experimental results of the MOLM for three benchmark tasks. **a** Schematic of the experimental setup of the MOLM for the benchmark tasks. **b** Image of the metasurface chip compared with a Hong Kong dollar coin. **c** Optical microscope image of the fabricated metasurface chip with a compact area of $3.2 \times 3.2 \text{ mm}^2$ (left graph) and the scanning electron microscope (SEM) image of a zoom-in metasurface region containing 25×25 meta-atoms (right graph). **d** The measurement result of quantitative phase imaging (QPI) of a zoom-in region of the metasurface chip. The details about the QPI measurement are provided in Supplementary Note 7. In the color wheel, the color represents the phase modulation coefficient, and the brightness represents the amplitude modulation coefficient. **e**, **h**, and **k** Illustration of the dataset images of the COVID-19 Radiography (**e**), the NIH ChestX-ray8 (**h**), and RSNA Intracranial Hemorrhage Detection tasks (**k**). **f** The accuracy versus optical neuron number of the MOLM for the COVID-19 Radiography. The dashed line represents the accuracy of SAM as a comparison. The inset graphs with colorful scatters represent the results of the t-SNE analysis. **i** The confusion matrix of the prediction result of the NIH ChestX-ray8. **l** Bleeding regions inside the brain predicted by the MOLM (bottom graph) and digital ResNet-50 (upper graph). **g**, **j**, and **m** Comparison of accuracy and electronic parameters of the MOLM with other optical approaches [76, 99] and digital models for the COVID-19 Radiography (**g**), the NIH ChestX-ray8 (**j**), and RSNA Intracranial Hemorrhage Detection tasks (**m**)

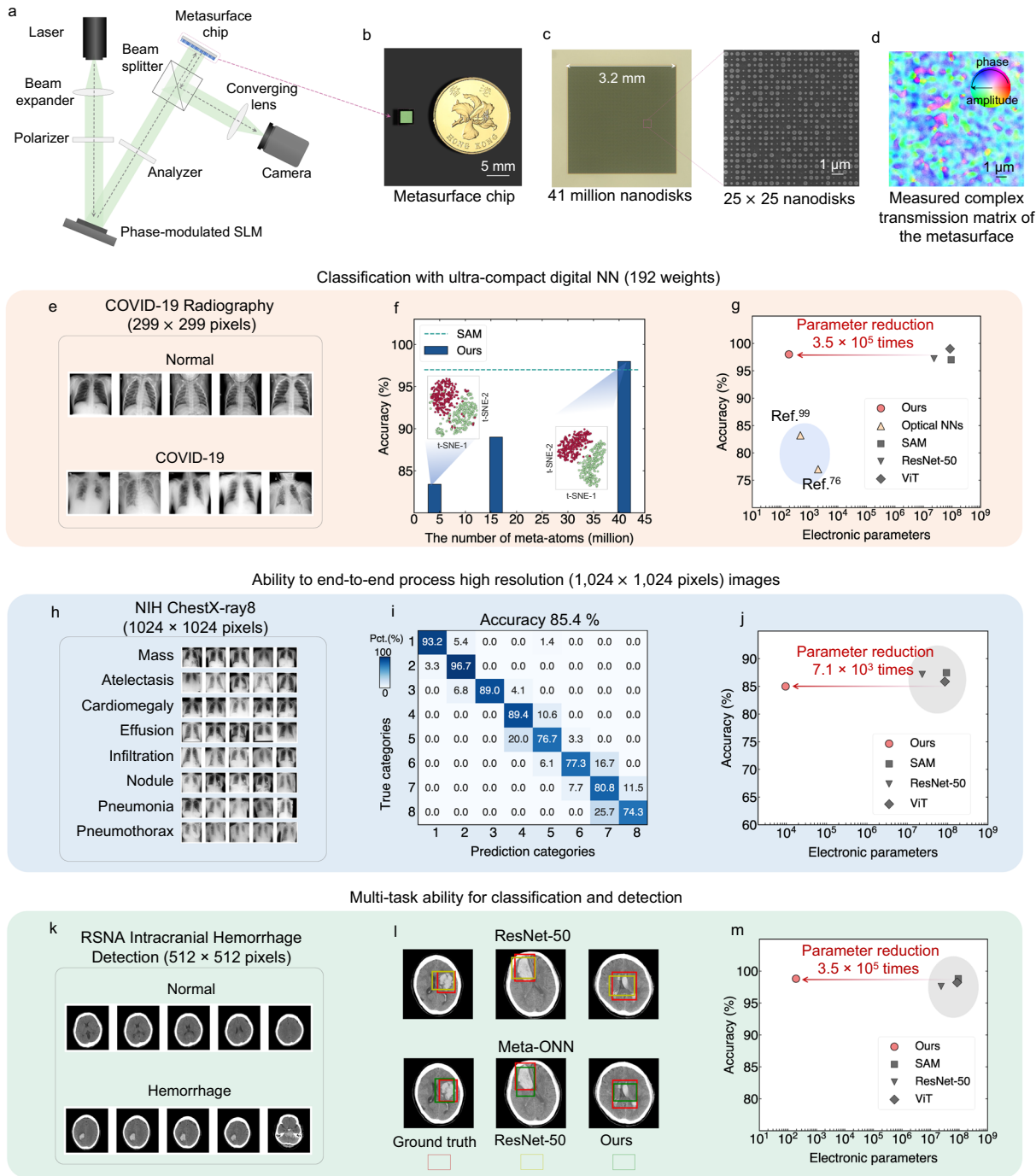


Fig. 2 (See legend on previous page.)

respectively. The energy consumption includes the energy consumed by the laser source, SLM, camera, and digital computer, respectively. More details about the time and energy consumption benchmarks can be found in Supplementary Notes 23 and 24.

Our investigation further shows that the substantial number of optical parameters unique to the metasurface device is the key factor contributing to the high accuracy. The accuracy increases from 83.0% to 98.0% as the number of optical parameters increases from 4.2 million

to 41 million, as shown in Fig. 2f (see details about experiments in Supplementary Note 9). To gain deeper insights into the data, we employ t-distributed stochastic neighbor embedding (t-SNE) [100] to visualize the data after being processed by the MOLM, where the original images are transformed into a lower-dimensional space by the metasurface. As shown in the inset graphs of Fig. 2f, the separation between different classes becomes highly distinct as the number of optical parameters increases substantially to 41 million.

Encouraged by the performance we have achieved, we delve into more challenging applications that have never been demonstrated by ONNs. These tasks showcase our MOLM's ability to process high-resolution images and tackle more challenging medical diagnostic tasks.

2.2.2 Capability for end-to-end processing of high-resolution images

Thoracic diseases detection: This task aims to demonstrate our system's ability to perform end-to-end processing of high-resolution images with over one million pixels. The NIH ChestX-Ray8 dataset [101] comprises high-resolution ($1,024 \times 1,024$ pixels) front-view X-ray images covering eight different thoracic diseases as shown in Fig. 2h. We demonstrate that these commonly occurring thoracic diseases can be detected by our MOLM with an average accuracy of 85.4% as shown in Fig. 2i, followed by minimal digital processing using 9,600 digitally trained weights. The achieved accuracy is comparable to results from ViT (85.8% accuracy) employing 85.8 million digitally trained weights. Additionally, this task highlights our MOLM's ability to process high-resolution images without any pre-processing, utilizing a pixel resolution of $1,024 \times 1,024$, the highest demonstrated in ONNs to date, to our best knowledge.

2.2.3 Multi-task ability for classification and detection

Intracranial Hemorrhage Detection: Intracranial hemorrhage (ICH) is a critical medical condition that requires rapid and intensive treatment to prevent further damage and to improve patient outcomes [102]. However, the timely detection of ICHs is often delayed due to a lack of prompt access to radiologists who read the scans. Here, we leverage our MOLM to expedite ICH detection in CT images, as shown in Fig. 2k. Our experiments solve two tasks. In the first task, we detect the existence of hemorrhages in CT images, achieving an accuracy of 98.8%, a precision of 98.5%, a sensitivity of 99.2%, and a specificity of 98.3%. In the second task, we identify the bleeding point location with the MOLM. We first downsample the optical images after MOLM to 15×20 pixels. These downsampled images serve as the input to a 6-layered fully connected neural network at the

digital backend. The output is a vector with 4 elements, each element representing the horizontal and vertical position coordinates, the width, and the height of a rectangular box. The box indicates the bleeding region in the CT images, as shown in Fig. 2l. Intersection over Union (IoU) is obtained from the predicted box and the true box to evaluate the performance of locating the bleeding region. Here, IoU is the ratio of the overlapped area over the united area of the two boxes. Our MOLM achieves an IoU of 0.61 for 3 different bleeding positions, meaning that the predicted bounding box indicating the bleeding location is well aligned with the ground truth box. In comparison, the average IoU obtained by ResNet-50 is 0.64.

An important discovery implied by the results of our MOLM across diverse vision tasks is its capacity of using only a single-layer metasurface combining a small-scale digital NN to compete against cutting-edge NN models with tens of millions of parameters. We experimentally validate in different tasks that our superior performance is credited to the massive parameters uniquely offered by the metasurface, as demonstrated in Fig. 2g, j, and m. While a small-scale NN is still necessary at the digital backend, its size can be highly compressed to up to a factor of 10^5 . Furthermore, it is worth noting that the remaining small NN at the backend can be readily implemented in analog devices, enabling the realization of an all-analog system.

2.3 Scalable to complex NN models for high-accuracy video sequence processing

Leveraging the computing framework of infinitely NNs built with random parameters, we can confidently increase the depth of the optical NN without concerns about physical errors accumulating layer by layer. This allows our system to tackle more complicated AI tasks. Here we construct a recurrent neural network (RNN) using our MOLM and apply it in video processing. Serving as a hidden layer, the MOLM is recurrently connected to form an RNN [40, 103]. This metasurface-based RNN (meta-RNN) takes a sequence of image frames from a video as input, as shown in Fig. 3a. Each frame is processed by the meta-RNN, with the output being downsampled and read out from the camera. Subsequently, this output is combined with the hidden state from previous frames to derive the hidden state of the current frame. Finally, the output is connected to a logistic regression-based classifier. The meta-RNN is employed for human action recognition tasks on the KTH dataset [104], which comprises six types of human actions across four different scenarios.

The evaluation of the performance of the model involves two types of action classification. One type identifies

actions jointly determined by sequences of frames, defined as action accuracy. The action is determined from the percentage of votes for all actions in each testing video sequence. The action with the maximum votes is the predicted action in a video sequence. The other type identifies the action indicated by each individual frame, defined as frame accuracy. With 9,180 trained weights, our meta-RNN obtains the frame accuracy of 97.5% and action accuracy of 99.1%, as shown in Fig. 3b and c, respectively. The training time to realize such a high accuracy is only 4.01 s using NVIDIA RTX 3090. The obtained accuracy exceeds those from digital NNs using long-short term memory (action accuracy of 90.7%) and the state-of-the-art ONNs (action accuracy of 96.3%). Additionally, unlike other ONNs that typically need preprocessing of the frames of the action sequences by pre-trained convolutional neural network (CNN) for human segmentation, our meta-RNN can directly process videos without the need for such preprocessing steps. Consequently, our meta-RNN is capable of processing videos at high speed, achieving 1,968 frames per second.

We further utilize this task to demonstrate the capability of our MOLM to rapidly recover from external perturbations and its potential for real-time adaptable learning. To show this property, we intentionally move the metasurface by 10 μm creating the axial misalignment after the digital NN is trained. The misalignment is the 20 times of the light wavelength. This axial misalignment results in a change in the modulation matrix of the metasurface, which may lead to a degradation of the accuracy by 16.7%. However, the MOLM is capable of quickly recovering from the error after re-training the single-layer digital NN as shown in Fig. 3e. The accuracy is restored to 96.3% after 45 epochs using only 1,920 training images, which takes only 234 ms. This study highlights the adaptability of the MOLM, even in dynamic or unpredictable environments and its potential for real-time adaptable learning. The experimental details can be found in Supplementary Note 10. Computing very large matrices with optical fan-in allows for extremely low optical energy consumption.

If the matrix size is sufficiently large, each multiplication requires a photon number far less than 1. Here, we demonstrate in Fig. 3f that an average photon number of 0.078 per multiplication is sufficient to maintain human action classification accuracy over 95%. This feature makes our system suitable for low-illumination environments. The experimental details can be found in Supplementary Note 11.

2.4 Solving real-world challenge: accelerating the analysis of multi-gigapixel whole slide images for cancer diagnosis

Leveraging on these advantages, we finally demonstrate the application of our MOLM in a real-world challenge, showcasing its distinct benefits in computationally intensive applications. For many diseases, particularly cancers, pathological diagnosis is the gold standard in clinical practice. The introduction of Whole Slide Imaging (WSI) scanners, which generate digitized pathology microscopic images, has revolutionized pathology image analysis. This technology enables computer-aided diagnostics, leveraging advanced deep-learning techniques to reduce the workload of pathologists and optimize the regional distribution of medical resources [107]. However, WSIs present a challenge for deep learning due to their extremely large size. With single slides containing multi-gigapixel images, efficient processing of these images is crucial for automated diagnosis.

In our work, we apply an MOLM to detect and localize breast cancer that has metastasized to nearby lymph nodes, a task of significant clinical importance but requiring substantial reading time from pathologists. Pathologists typically need to review thousands of megapixel photos from a single WSI exceeding 10 gigapixels. We use the CAMELYON16 dataset [108] for our study. To address the processing of extensive WSIs, each with dimensions of more than 2 billion pixels, we employ a patch-based framework. Initially, a preprocessing algorithm, Otsu algorithm [109], is used to separate the raw

(See figure on next page.)

Fig. 3 Experimental results of the meta-RNN for video-based human action recognition. **a** The working flow of the meta-RNN for video-based human action recognition. The action video is processed by the meta-RNN in a frame-by-frame way. Each action video contains 20 frames with a time interval of 80 ms. Firstly, the input frame x_t with 120×160 pixels is encoded into the optical domain without any preprocessing. It is then processed by the MOLM to generate a compressed digital image $f(x_t)$ with 34×45 pixels. The processed frame is fused with the output at the previous frame to form the output at the current frame, i.e., $h_t = h_{t-1} + f(x_t)$. Finally, the output at the current frame is fed into a digital neural network to produce a frame-wise prediction result y_t . The action prediction result is the label that appears most frequently in all frame-wise results of the action video. **b** The confusion matrix of frame-wise prediction results. **c** The confusion matrix of action prediction results. **d** Performance comparison of the MOLM with the state-of-the-art digital and optical approaches [40, 68, 105, 106]. The training time and processing speed of the meta-RNN are the predicted ones based on current experimental results (see details in Supplementary Note 23). **e** The ability of the meta-RNN in quickly recovering the accuracy after experiencing a hard perturbation. **f** The action accuracy versus averaged photons per complex-valued multiplication

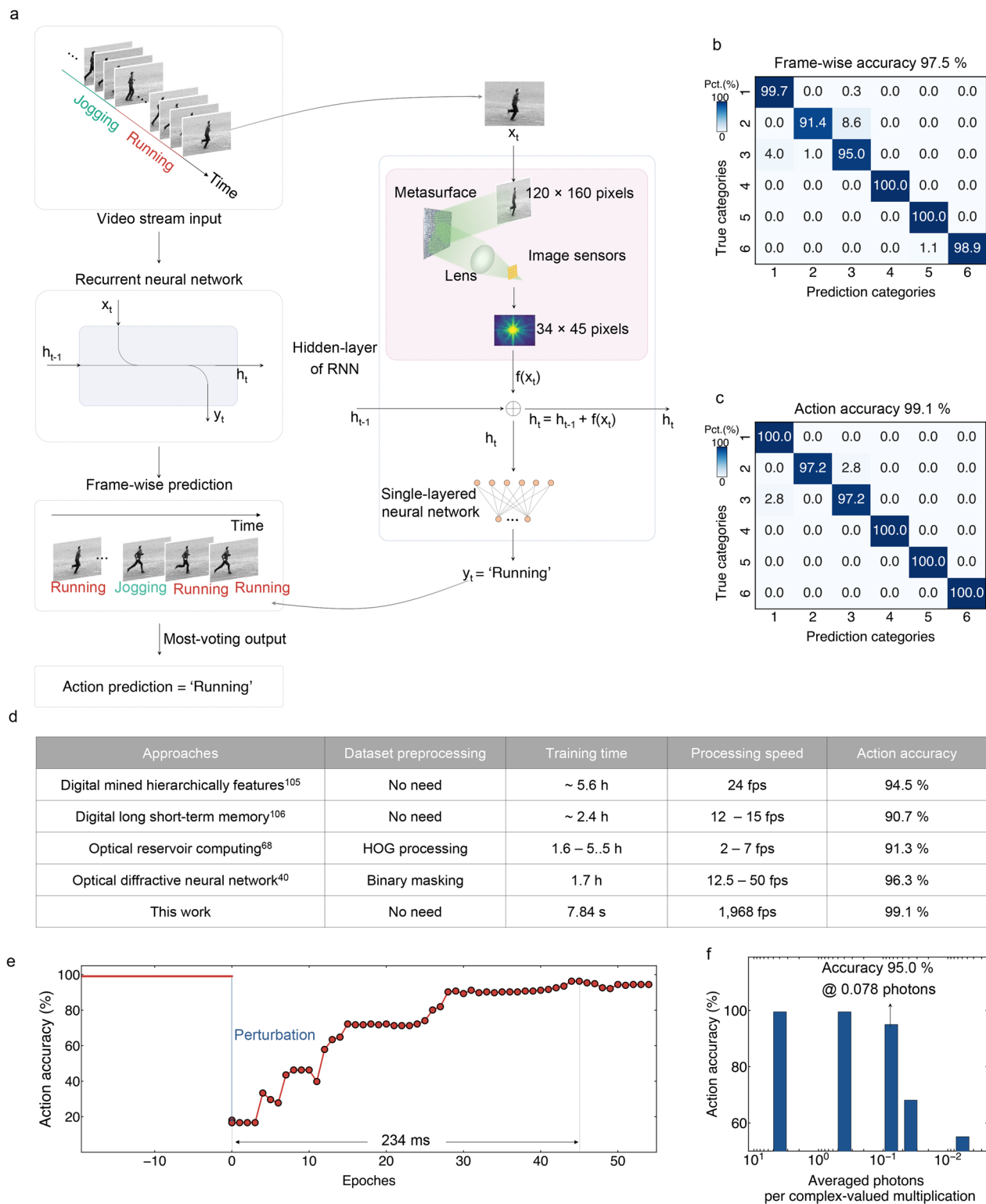


Fig. 3 (See legend on previous page.)

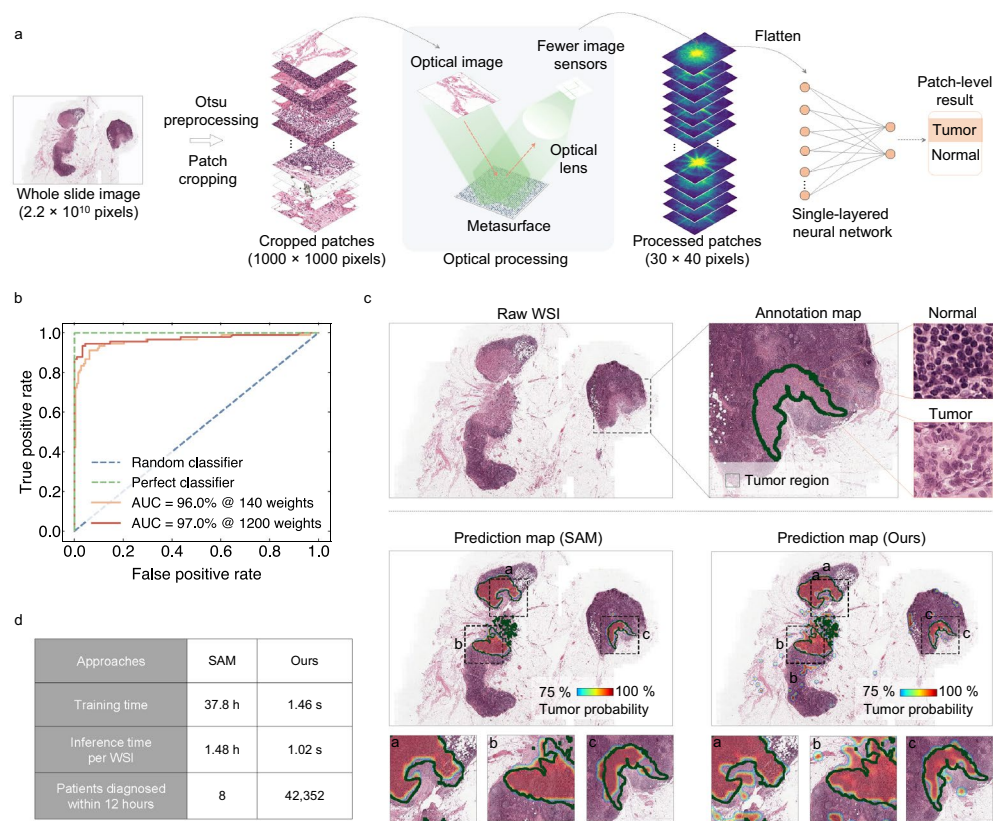


Fig. 4 Experimental results of the MOLM for cancer diagnosis based on the whole slide image. **a** The working flow of the cancer diagnosis based on the whole slide image. The input WSI with over 2.2×10^{10} pixels is preprocessed with the Otsu method and cropped into a series of patch images with $1,000 \times 1,000$ pixels. The patch images are processed by the MOLM in a single shot and compressed into 30×40 pixels. Finally, these processed images are fed to a highly compact digital neural network to generate the final prediction result of whether the tumor cell is detected in the image. **b** ROC curve of the patch-level prediction results during the training phase. **c** The heat map of the prediction probabilities of the MOLM and the state-of-the-art segmentation model (SAM). The inset graphs a, b, and c represent three different zoom-in regions of the WSI, respectively. **d** Comparison of the training time and inference time of the MOLM with SAM. The time consumption of the MOLM is the predicted one based on current experimental results (see the details in Supplementary Note 23)

whole slide image into the useful foreground and the non-tissue background, resulting in a reduced total pixel of the whole slide image to be processed, as shown in Fig. 4a. We then divide the large WSIs into smaller patches, each containing $1,000 \times 1,000$ pixels. These patches consist of 1,775 normal patches and 887 tumor patches. During the training phase, we convert the patches into optical images using the SLM. The modulated patch samples are processed by our MOLM and detected by the sensor array. Subsequently, the output from the sensor array is downsampled. The final step is to train a single-layered neural network with the patch samples to create a classifier capable of distinguishing between normal and tumor classes. The mean AUC is 96.0% when the trained weight number is 140 and increases to 97.0% at the weight number of 1,200, as shown in Fig. 4b. The training process takes only 1.46 s, achieving a training accuracy of 95.1%, as shown in Fig. 4d.

Another WSI consisting of 2,030 unlabeled patches is used to test the performance of tumor tissue segmentation. Initially, these unlabeled patches are first processed using our MOLM. Subsequently, the processed patches are inferred with the trained single-layered NN to produce the tumor-positive probability. Finally, the probability heat map is generated by mapping the predicted probability of the patches to the raw WSI. As shown in Fig. 4c, the resulting heat map demonstrates that our MOLM achieves accurate segmentation of three different tumor tissue regions from the billion-pixel-scale WSI, achieving an IOU of 0.60, which is comparable to that of SAM (0.63). More importantly, our MOLM exhibits an impressively fast inference time of 1.02 s per whole slide image (WSI), representing a significant reduction compared to SAM which requires 1.48 h to analyze one WSI. This remarkable reduction in inference time allows our

MOLM to diagnose more than 42,352 patients within a single 12-h working day, while only 8 patients can be diagnosed using SAM in the same timeframe.

3 Discussions

Connections and differences with digital ELM: The overall architecture is equivalent to a standard ELM, but with two key differences: our hidden-layer nonlinearity is quadratic rather than sigmoidal, and the fully connected weights are complex-valued and modulated by diffraction rather than arbitrary real-valued random variables. In Supplementary Note 5, we show that our system with complex-valued parameters slightly outperforms a digital ELM as the hidden-layer width increases; however, both exceed 99% accuracy on MNIST when the parameter count reaches tens of millions.

Operation under incoherent light illumination: One advantage of the meta-ONN is its compatibility with broadband illumination, which greatly expands the application scope of our system. Although metasurface responses are wavelength dependent, our approach does not rely on precisely optimized optical parameters, so spectral variations have a limited impact on performance. To experimentally demonstrate this, a broadband continuous LED source is combined with wavelength band-pass filters to realize different spectral widths from 20 nm to 100 nm. The LED light contains all wavelengths within the chosen spectral band. We demonstrate that MNIST inference can achieve >98.3% accuracy across all spectral widths. The modest drop relative to 532-nm monochromatic laser operation (99.2%) arises because LED-based optics implement only intensity-domain linear operations, whereas coherent light enables complex-valued linear transformations, yielding higher model capacity. The experimental details are provided in Supplementary Note 12.

Fabrication variances tolerance: Typical optical metasurfaces are highly sensitive to fabrication variations due to their subwavelength feature sizes; even with the same design, performance can vary substantially across devices. In contrast, our system is much more robust because it does not rely on the precise realization of individual meta-atoms. To demonstrate this, we tested three chips fabricated in different batches. The meta-atoms of the fabricated metasurface exhibit a diameter range from 92 nm to 413 nm, deviating from the designed diameter range from 100 nm to 400 nm due to fabrication errors. Despite the existence of physical imperfections, we observe that the experimental MNIST accuracy varies by less than 1% across the devices. The experimental results are provided in Methods and Supplementary Note 4.

Comparison with random scattering media: Random scattering media, which strongly modulate

optical amplitude and phase, could also serve as an optical frontend in our architecture. The key distinction is controllability: random media are defined by micrometre-scale scatterers, whereas metasurfaces comprise engineered nanometre-scale meta-atoms, providing a far higher density of effective modulation elements. This finer spatial control enables more efficient spatial information encoding and, consequently, higher classification accuracy. Experimentally, replacing the metasurface with a random scattering medium yields 92.0% accuracy on MNIST, substantially lower than the 99.2% achieved with the metasurface. Experimental details are provided in Supplementary Note 13.

Improving system compactness: The sub-wavelength optical metasurface is much thinner and smaller than bulk optical devices. This allows for a highly compact integration with the CMOS sensors and thus a minimized system volume of our MOLM system. The system volume of a free-space-based ONN is defined as the area of the free-space optical device multiplied by the spacing distance between the optical device and the COMS camera. In the current experiment setup which uses a reflective metasurface, the reduction of volume is hindered by the use of a beam splitter, resulting in a system volume of 2,560 mm³. In an improved design, the metasurface could be designed to work in a transmissive way by using a transparent substrate, for example, sapphire and quartz. The beam splitter is not needed. The distance between the metasurface and the converging lens could be shortened. In addition, the bulky converging lens could be replaced with a thin metalens with a high numerical aperture of 0.99 [110]. Considering that the metalens needs to fully cover the metasurface with a squared area of 3.2×3.2 mm², the diameter of the metalens is 4.5 mm and the focal length is 2.27 mm. The minimum distance between the metasurface and the lens can be equal to the metasurface size of 3.2 mm, which enables the all-to-all connection of optical weights during light diffraction. The spacing between the metasurface and the camera can be as small as 5.47 mm. As a result, the system volume could be extremely shrunk to 56 mm³, which is more compact by 10^2 – 10^4 times than most state-of-the-art 3D free-space-based ONN systems [33, 39, 40, 42]. This compact volume allows the optical system to be fully integrated with the following digital backend.

Possible limitations: Compared with digital NN, the current system exhibits two limitations that can improvement when addressing more complex tasks. First, the architecture of the digital backend is relatively simple, consisting only of fully connected layers, whereas state-of-the-art digital ELM typically employ more advanced architectures such as deep convolutional NNs and transformers. The simple architecture of the

digital backend would result in inadequate learning ability and thus unsatisfactory performance in complex tasks. Second, we use an optical lens that transforms image into the Fourier domain. We found images in the Fourier domain exhibit obvious frequency-related features that are useful for classification tasks. However, for tasks requiring pixel-wise prediction, such as image segmentation, the optical lens will degrade system performance. In future work, these limitations could be overcome by designing a task-specific digital backend and removing the optical lens from the system.

In summary, we experimentally demonstrate a high-performance MOLM that provides unprecedented accuracy and scalability. These exceptional performances are a result of realizing a true large-scale optical computing system by innovatively combining optical metasurfaces, a device fully leveraging the parallelism of optics, with infinitely wide NNs, a computing framework suitable for metasurface-based NNs to scale to arbitrary widths, depths, and complexity. An important observation from our experimental demonstration is that a single-layer optical metasurface chip with 41 million disorder-engineered meta-atoms, combining a small-scale digital NN, can achieve competitive performance compared to state-of-the-art deep NNs such as ResNet and ViT. Our work represents a significant advancement, moving ONNs beyond benchmark demonstrations to effectively address real-world challenges. The performance comparison of the MOLM with the state-of-the-art ONN systems is shown in Table S8 in Supplementary Note 27.

Our system loads over 99.99% computations on a passive optical metasurface. Although a small digital NN is still required at the backend, the time for training such as NN is greatly shortened from tens of hours to a few seconds. If high-speed SLMs and cameras are used, the training time could be reduced to only tens of milliseconds. This makes our system highly adaptable to dynamic or unpredictable environments that require real-time learning abilities, such as robot-assisted surgery. Meanwhile, the inference time could easily be improved to the microsecond or even sub-microsecond level, which could enable a range of applications involving high-speed objects and requiring real-time decisions. Even shorter training and inference times could be achieved using delicate neural network accelerators realized with field-programmable gate arrays (FPGAs) and all-analog electronics [111, 112]. Overall, our system provides a practical pathway toward high-performance optical AI computing, enabled by its scalability, high tolerance to fabrication and alignment errors, and ability to operate under both coherent and incoherent illumination.

4 Methods

Metasurface design and simulation: The metasurface is created using massive periodic unit cells. Every unit cell consists of a 220-nm-thick silicon cylindrical meta-atoms with a diameter that varies across different cells. The silicon meta-atoms is deposited on a 2- μm -thick buried oxide layer on a 725- μm -thick silicon substrate. These unit cells are arranged in a square lattice in the two-dimensional plane. The period of the metasurface unit cell is 500 nm. Based on three-dimensional FDTD simulations, the optical phase and amplitude modulation coefficients of the unit cell for the incident light are obtained at various diameters under normal incidence. In the simulations, the grid size is 0.25 nm for the meta-atom and 8 nm for the buried oxide and substrate layers, respectively. The boundaries at the x - y plane are periodic to emulate the periodic meta-atoms of the metasurface, and the boundaries at the z plane are perfect match layers. The simulation details are provided in Supplementary Note 1. Simulation results indicates that the optical phase and amplitude of the incident light can be effectively manipulated by varying the diameter of the meta-atoms. The data table for the mapping relationship between meta-atom diameter and phase modulation is shown in Figure S1c of Supplementary Information. The fabricated metasurface comprises $6400 \times 6400 = 41$ million meta-atoms. The diameter of every meta-atom is independently designed to realize the optical phase and amplitude modulation coefficients following a Gaussian sampling distribution. Specifically, the standard deviation of the Gaussian distribution of phase modulation is designed as 0.4π . With the designed phase distribution, the diameter of every meta-atom can be obtained according to the mapping relationship between the optical phase and meta-atom diameter, as shown in Figure S1c of Supplementary Information. The range of the meta-atom diameter is from 100 nm to 400 nm. The amplitude distribution is also determined without the need for further designs according to the relationship between optical amplitude and meta-atom's diameter.

Metasurface fabrication: The metasurface chip is fabricated based on a multi-project-wafer (MPW) process offered by a commercial electron-beam-lithography-based silicon foundry. The fabrication process starts with a silicon-on-insulator (SOI) wafer, where the device layer thickness is 220 nm, the buffer oxide layer thickness is 2 μm , and the silicon substrate thickness is 725 μm . A layer of electron-sensitive resist is coated on the device layer of the wafer, followed by heating to strengthen the resist. After that, a 100 keV electron gun is used to define the meta-atom patterns of the metasurface. The wafer is then chemically developed, and the patterned resist remains on the substrate. Finally,

an anisotropic RIE etching process is performed to transfer the pattern of the resist into the silicon device layer. The etching process will continue until the surface of the buffer oxide layer is exposed.

Quantitative phase imaging of the fabricated metasurface: Quantitative Phase Imaging (QPI) is used to measure the reflective phase response of the metasurface. The system has a diffraction-limited spatial resolution of 665 nm and a field of view of $36 \times 36 \mu\text{m}^2$. With a total magnification of $200\times$, the system achieves a high pixel resolution of 60 nm, thus enabling a clear visualization of the phase modulation matrix. The details about the QPI measurement are provided in Supplementary Note 7.

Experimental setup of the metasurface-based optical learning machine (MOLM): Our MOLM comprises an SLM for optical image generation, a single-layer metasurface chip, a converging lens for coherent summation, and a camera for optical image capture. The experiment setup starts with a 532-nm laser diode which emits an optical beam with a diameter of 3 mm and a spectrum with a full width at half maximum (FWHM) of 10 nm. The laser is then collimated using an optical lens and expanded to a diameter of 10 mm using an optical 4-f system. The purpose of expanding the beam diameter is to ensure that the laser beam can effectively cover the active area ($15.4 \text{ mm} \times 9.6 \text{ mm}$) of the spatial light modulator (SLM, Meadowlark Optics E19x12-400–800-HDM8). The SLM has a full-pixel resolution of $1,920 \times 1,200$, a pitch size of $8 \mu\text{m}$, and an 8-bit modulation depth. The expanded laser beam enters onto the SLM at an off-axis angle of 12° . By introducing two linear polarizers placed before and after the SLM, the SLM can be used to encode the digital image onto the optical domain, as introduced in Supplementary Note 4. The polarization angle of both polarizers is 45° with respect to the horizontal direction. The SLM-generated optical image is projected onto the metasurface chip. Here, to ensure that the optical image can effectively overlap with the area of the metasurface chip to realize full-area modulation of the optical image, the effective area of the optical image is intentionally set as $4.1 \times 4.1 \text{ mm}^2$. Two approaches are used to realize the consistent effective area in all the tasks: (1) For the task images whose pixel scale is smaller than 512×512 , the images are upsampled to a pixel scale of 512×512 using the nearest area interpolation method in the digital domain. Consequently, the effective area of the optical image is $512 \times 8 \mu\text{m} \times 512 \times 8 \mu\text{m} = 4.1 \times 4.1 \text{ mm}^2$. For example, the pixel scale of the COVID-19 image is upsampled to be 512×512 from 299×299 . (2) For the task images

whose pixel scale is larger than 512×512 , the generated optical image is minified to an effective area of $4.1 \times 4.1 \text{ mm}^2$ using a 4-f system in the optical domain. For example, the Chest X-ray image with a pixel scale of $1,024 \times 1,024$ is minified double. In addition, since the effective area of the optical image is smaller than the laser beam size ($10 \times 10 \text{ mm}^2$), an aperture of $5.5 \times 5.5 \text{ mm}^2$ is used to block the stray light of the laser beam. A 50/50 beam splitter is placed between the SLM and the metasurface chip to realize the normal incidence of the optical image to the metasurface. The optical image is reflected by the metasurface and the beam splitter in sequence. Following that, the reflected optical image is focused by an optical converging lens with a focal length of 150 mm. Lastly, a camera (IRVI Contour IR Digital CMOS Camera) with an 8-bit depth and an activated pixel number of 600×800 is placed close to the focal plane of the converging lens to capture the optical image. The figure showing the experimental setup is shown in Figure S4 of Supplementary Information.

Neural network training: In the experiment, the camera-captured images processed by the MOLM are fed into a single-layer backend digital neural network to produce the prediction results of various machine vision tasks. The working flow of neural network training is shown in Figure S14a of Supplementary Information. First, the captured images are downsampled by averaging the received power in the neighboring sensor array, see the details in Figure S14b of Supplementary Information. After that, the down-sampled images and the corresponding labels are combined as the new dataset after the MOLM. The dataset is then randomly split into the training dataset and testing dataset by a certain ratio. The training dataset is used to train the digital neural network. The digital neural network consists of an input layer for receiving the input image and an output layer for generating prediction results, and the two layers are fully connected, as shown in Figure S14c of Supplementary Information. Biases are added to the output before the activation function of *sigmoid*. The mathematical expression of the single-layered digital neural network is $y = \text{sigmoid}(b_0 + w_0x)$, where b_0 , w_0 , x , and y are the biases, the digital weights, the input image in the form of vector and the prediction result, respectively. Logistic regression is used to optimize the biases and digital weights. The nonlinear activation function is *sigmoid*. Finally, the testing dataset is used to verify the performance of the overall system. The downsampling ratio is 1,600 for MNIST, 676 for CIFAR-10, 2,500 for COVID-19,

2,500 for the ICH dataset, 400 for the NIH dataset, 400 for the WSI cancer dataset, and 324 for the human action recognition, respectively. The split ratio of the dataset and the hyperparameters for neural network training are task-dependent, which are introduced in Supplementary Notes 11–17. The details about the digital model training (SAM, ResNet-50, and ViT) are provided in Note 22.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s43593-026-00127-y>.

Supplementary material 1.

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Author contributions

C.H. and M.L. conceived the idea. M.L. designed the metasurface chip, conducted the experiments, and analyzed the experimental data. M.L., J.X., and W.G. conducted the optical simulations. M.J. conducted experiments for digital neural networks. N.Z. conducted the QPI experiment. R.Z. supervised the QPI experiment. C.H. led the writing of the main manuscript. M.L. and B.S., M.J., C.S., and Q.D. contributed to the paper writing. C.H., Q.D., and C.S. supervised the project.

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests

Chaoran Huang, Chester Shu, and Mingcheng Luo declare US provisional patent application number 63/643,973.

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References

1. Y. LeCun, Y. Bengio, G. Hinton, Deep learning. *Nature* **521**(7553), 436–444 (2015)
2. J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F.L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al. Gpt-4 technical report (2023), arXiv preprint [arXiv:2303.08774](https://arxiv.org/abs/2303.08774)
3. B.J. Shastri, A.N. Tait, T. Lima, W.H. Pernice, H. Bhaskaran, C.D. Wright, P.R. Prucnal, Photonics for artificial intelligence and neuromorphic computing. *Nat. Photonics* **15**(2), 102–114 (2021)
4. G. Wetzstein, A. Ozcan, S. Gigan, S. Fan, D. Englund, M. Soljačić, C. Denz, D.A. Miller, D. Psaltis, Inference in artificial intelligence with deep optics and photonics. *Nature* **588**(7836), 39–47 (2020)
5. P.L. McMahon, The physics of optical computing. *Nat. Rev. Phys.* **5**(12), 717–734 (2023)
6. N.H. Farhat, D. Psaltis, A. Prata, E. Paek, Optical implementation of the hopfield model. *Appl. Opt.* **24**(10), 1469–1475 (1985)
7. W. Bogaerts, D. Pérez, J. Capmany, D.A. Miller, J. Poon, D. Englund, F. Morichetti, A. Melloni, Programmable photonic circuits. *Nature* **586**(7828), 207–216 (2020)
8. J. Sun, E. Timurdogan, A. Yaacobi, E.S. Hosseini, M.R. Watts, Large-scale nanophotonic phased array. *Nature* **493**(7431), 195–199 (2013)
9. X. Zhang, K. Kwon, J. Henriksson, J. Luo, M.C. Wu, A large-scale microelectromechanical-systems-based silicon photonics lidar. *Nature* **603**(7900), 253–258 (2022)
10. Y. Shen, N.C. Harris, S. Skirlo, M. Prabhu, T. Baehr-Jones, M. Hochberg, X. Sun, S. Zhao, H. Larochelle, D. Englund, Deep learning with coherent nanophotonic circuits. *Nat. Photonics* **11**(7), 441–446 (2017)
11. P. Stark, F. Horst, R. Dangel, J. Weiss, B.J. Offrein, Opportunities for integrated photonic neural networks. *Nanophotonics* **9**(13), 4221–4232 (2020)
12. C. Pappas, T. Moschos, M. Moralis-Pegios, G. Giamougiannis, A. Tsakyridis, M. Kirtas, N. Passalis, A. Tefas, N. Pleros, A teraflop photonic matrix multiplier using time-space-wavelength multiplexed awgr-based architectures. In: 2024 Optical Fiber Communications Conference and Exhibition (OFC), (IEEE, 2024), p. 1–3
13. L. De Marinis, M. Cocconcioni, O. Liboiron-Ladouceur, G. Contestabile, P. Castoldi, N. Andriolli, Photonic integrated reconfigurable linear processors as neural network accelerators. *Appl. Sci.* **11**(13), 6232 (2021)
14. G. Donati, C.R. Mirasso, M. Mancinelli, L. Pavesi, A. Argyris, Microring resonators with external optical feedback for time delay reservoir computing. *Opt. Express* **30**(1), 522–537 (2021)
15. A. Skalli, J. Robertson, D. Owen-Newns, M. Hejda, X. Porte, S. Reitzenstein, A. Hurtado, D. Brunner, Photonic neuromorphic computing using vertical cavity semiconductor lasers. *Opt. Mater. Express* **12**(6), 2395–2414 (2022)
16. K. Sozos, A. Bogris, P. Bienstman, G. Sarantoglou, S. Deligiannidis, C. Mesaritakis, High-speed photonic neuromorphic computing using recurrent optical spectrum slicing neural networks. *Commun. Eng.* **1**(1), 24 (2022)
17. R. Amin, J.K. George, H. Wang, R. Maiti, Z. Ma, H. Dalir, J.B. Khurgin, V.J. Sorger, An ITO-graphene heterojunction integrated absorption modulator on Si-photonics for neuromorphic nonlinear activation. *Appl. Photon.* **6**(12), 120801 (2021)
18. X. Xu, M. Tan, B. Corcoran, J. Wu, A. Boes, T.G. Nguyen, S.T. Chu, B.E. Little, D.G. Hicks, R. Morandotti, 11 tops photonic convolutional accelerator for optical neural networks. *Nature* **589**(7840), 44–51 (2021)
19. B. Shi, N. Calabretta, R. Stabile, Inp photonic integrated multi-layer neural networks: architecture and performance analysis. *APL Photon.* **7**(1), 010801 (2022)
20. C. Wu, H. Yu, S. Lee, R. Peng, I. Takeuchi, M. Li, Programmable phase-change metasurfaces on waveguides for multimode photonic convolutional neural network. *Nat. Commun.* **12**(1), 96 (2021)
21. B. Dong, S. Aggarwal, W. Zhou, U.E. Ali, N. Farmakidis, J.S. Lee, Y. He, X. Li, D.-L. Kwong, C. Wright, Higher-dimensional processing using a photonic tensor core with continuous-time data. *Nat. Photon.* **17**(12), 1080–1088 (2023)
22. J.M. Shainline, S.M. Buckley, R.P. Mirin, S.W. Nam, Superconducting optoelectronic circuits for neuromorphic computing. *Phys. Rev. Appl.* **7**(3), 034013 (2017)
23. X. Xiao, M.B. On, T. Van Vaerenbergh, D. Liang, R.G. Beausoleil, S. Yoo, Large-scale and energy-efficient tensorized optical neural networks on iii-v-on-silicon moscap platform. *APL Photon.* **6**(12), 126107 (2021)
24. T. Fu, Y. Zang, Y. Huang, Z. Du, H. Huang, C. Hu, M. Chen, S. Yang, H. Chen, Photonic machine learning with on-chip diffractive optics. *Nat. Commun.* **14**(1), 70 (2023)
25. J. Cheng, Y. Zhao, W. Zhang, H. Zhou, D. Huang, Q. Zhu, Y. Guo, B. Xu, J. Dong, X. Zhang, A small microring array that performs large complex-valued matrix-vector multiplication. *Front. Optoelectron.* **15**(1), 15 (2022)
26. B. Bai, Q. Yang, H. Shu, L. Chang, F. Yang, B. Shen, Z. Tao, J. Wang, S. Xu, W. Xie, Microcomb-based integrated photonic processing unit. *Nat. Commun.* **14**(1), 66 (2023)
27. S. Xu, J. Wang, S. Yi, W. Zou, High-order tensor flow processing using integrated photonic circuits. *Nat. Commun.* **13**(1), 7970 (2022)

28. X. Meng, G. Zhang, N. Shi, G. Li, J. Azaña, J. Capmany, J. Yao, Y. Shen, W. Li, N. Zhu, Compact optical convolution processing unit based on multimode interference. *Nat. Commun.* **14**(1), 3000 (2023)
29. F. Ashtiani, A.J. Geers, F. Aflatouni, An on-chip photonic deep neural network for image classification. *Nature* **606**(7914), 501–506 (2022)
30. J. Feldmann, N. Youngblood, M. Karpov, H. Gehring, X. Li, M. Stappers, M. Le Gallo, X. Fu, A. Lukashchuk, A.S. Raja, Parallel convolutional processing using an integrated photonic tensor core. *Nature* **589**(7840), 52–58 (2021)
31. N.U. Dinc, A. Saba, J. Madrid-Wolff, C. Gigli, A. Boniface, C. Moser, D. Psaltis, From 3d to 2d and back again. *Nanophotonics* **12**(5), 777–793 (2023)
32. M. Yildirim, N.U. Dinc, I. Oguz, D. Psaltis, C. Moser, Nonlinear processing with linear optics. *Nat. Photonics* **18**(10), 1076–1082 (2024)
33. T. Wang, M.M. Sohoni, L.G. Wright, M.M. Stein, S.-Y. Ma, T. Onodera, M.G. Anderson, P.L. McMahon, Image sensing with multilayer nonlinear optical neural networks. *Nat. Photonics* **17**(5), 408–415 (2023)
34. J. Chang, V. Sitzmann, X. Dun, W. Heidrich, G. Wetzstein, Hybrid optical-electronic convolutional neural networks with optimized diffractive optics for image classification. *Sci. Rep.* **8**(1), 1–10 (2018)
35. F. Xia, K. Kim, Y. Eliezer, S. Han, L. Shaughnessy, S. Gigan, H. Cao, Nonlinear optical encoding enabled by recurrent linear scattering. *Nat. Photonics* **18**(10), 1067–1075 (2024)
36. K. Wei, X. Li, J. Froech, P. Chakravarthula, J. Whitehead, E. Tseng, A. Majumdar, F. Heide, Spatially varying nanophotonic neural networks. *Sci. Adv.* **10**(45), 0391 (2024)
37. C. Liu, Q. Ma, Z.J. Luo, Q.R. Hong, Q. Xiao, H.C. Zhang, L. Miao, W.M. Yu, Q. Cheng, L. Li, A programmable diffractive deep neural network based on a digital-coding metasurface array. *Nat. Electron.* **5**(2), 113–122 (2022)
38. M. Luo, T. Xu, S. Xiao, H.K. Tsang, C. Shu, C. Huang, Meta-optics based parallel convolutional processing for neural network accelerator. *Laser Photon. Rev.* **18**(11), 2300984 (2024)
39. X. Lin, Y. Rivenson, N.T. Yardimci, M. Veli, Y. Luo, M. Jarrahi, A. Ozcan, All-optical machine learning using diffractive deep neural networks. *Science* **361**(6406), 1004–1008 (2018)
40. T. Zhou, X. Lin, J. Wu, Y. Chen, H. Xie, Y. Li, J. Fan, H. Wu, L. Fang, Q. Dai, Large-scale neuromorphic optoelectronic computing with a reconfigurable diffractive processing unit. *Nat. Photonics* **15**(5), 367–373 (2021)
41. E. Goi, X. Chen, Q. Zhang, B.P. Cumming, S. Schoenhardt, H. Luan, M. Gu, Nanoprinted high-neuron-density optical linear perceptrons performing near-infrared inference on a cmos chip. *Light Sci. Appl.* **10**(1), 40 (2021)
42. X. Luo, Y. Hu, X. Ou, X. Li, J. Lai, N. Liu, X. Cheng, A. Pan, H. Duan, Metasurface-enabled on-chip multiplexed diffractive neural networks in the visible. *Light Sci. Appl.* **11**(1), 158 (2022)
43. H. Zheng, Q. Liu, Y. Zhou, I.I. Kravchenko, Y. Huo, J. Valentine, Meta-optic accelerators for object classifiers. *Sci. Adv.* **8**(30), 6410 (2022)
44. H. Zheng, Q. Liu, I.I. Kravchenko, X. Zhang, Y. Huo, J.G. Valentine, Multi-channel meta-imagers for accelerating machine vision. *Nat. Nanotechnol.* (2024). <https://doi.org/10.1038/s41565-023-01557-2>
45. S.-Y. Ma, T. Wang, J. Laydevant, L.G. Wright, P.L. McMahon, Quantum-noise-limited optical neural networks operating at a few quanta per activation. *Res. Sq.* (2023). <https://doi.org/10.1038/s41467-024-55220-y>
46. T. Wang, S.-Y. Ma, L.G. Wright, T. Onodera, B.C. Richard, P.L. McMahon, An optical neural network using less than 1 photon per multiplication. *Nat. Commun.* **13**(1), 123 (2022)
47. A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need. *Advances in neural information processing systems*, vol. 30 (2017)
48. A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale (2020), arXiv preprint [arXiv:2010.11929](https://arxiv.org/abs/2010.11929)
49. V.-C. Su, C.H. Chu, G. Sun, D.P. Tsai, Advances in optical metasurfaces: fabrication and applications. *Opt. Express* **26**(10), 13148–13182 (2018)
50. W.T. Chen, A.Y. Zhu, F. Capasso, Flat optics with dispersion-engineered metasurfaces. *Nat. Rev. Mater.* **5**(8), 604–620 (2020)
51. D. Neshev, I. Aharonovich, Optical metasurfaces: new generation building blocks for multi-functional optics. *Light Sci. Appl.* **7**(1), 58 (2018)
52. Y. Zhao, X.-X. Liu, A. Alù, Recent advances on optical metasurfaces. *J. Opt.* **16**(12), 123001 (2014)
53. S. Chen, W. Liu, Z. Li, H. Cheng, J. Tian, Metasurface-empowered optical multiplexing and multifunction. *Adv. Mater.* **32**(3), 1805912 (2020)
54. X. Qiu, J. Zhang, Y. Fan, J. Zhou, L. Chen, D.P. Tsai, Metasurface enabled high-order differentiator. *Nat. Commun.* **16**(1), 2437 (2025)
55. Y. Zhou, H. Zheng, I.I. Kravchenko, J. Valentine, Flat optics for image differentiation. *Nat. Photonics* **14**(5), 316–323 (2020)
56. G. Qu, G. Cai, X. Sha, Q. Chen, J. Cheng, Y. Zhang, J. Han, Q. Song, S. Xiao, All-dielectric metasurface empowered optical-electronic hybrid neural networks. *Laser & Photonics Reviews* **16**(10), 2100732 (2022)
57. D. Mengü, Y. Zhao, N.T. Yardimci, Y. Rivenson, M. Jarrahi, A. Ozcan, Misalignment resilient diffractive optical networks. *Nanophotonics* **9**(13), 4207–4219 (2020)
58. Y. Luo, D. Mengü, N.T. Yardimci, Y. Rivenson, M. Veli, M. Jarrahi, A. Ozcan, Design of task-specific optical systems using broadband diffractive neural networks. *Light Sci. Appl.* **8**(1), 112 (2019)
59. J. Dong, M. Rafayelyan, F. Krzakala, S. Gigan, Optical reservoir computing using multiple light scattering for chaotic systems prediction. *IEEE J. Sel. Top. Quantum Electron.* **26**(1), 1–12 (2019)
60. U. Paudel, M. Luengo-Kovac, J. Pilawa, T.J. Shaw, G.C. Valley, Classification of time-domain waveforms using a speckle-based optical reservoir computer. *Opt. Express* **28**(2), 1225–1237 (2020)
61. P. Antonik, N. Marsal, D. Rontani, Large-scale spatiotemporal photonic reservoir computer for image classification. *IEEE J. Sel. Top. Quantum Electron.* **26**(1), 1–12 (2019)
62. Y. Paquot, F. Duport, A. Smerieri, J. Dambre, B. Schrauwen, M. Haelterman, S. Massar, Optoelectronic reservoir computing. *Sci. Rep.* **2**(1), 287 (2012)
63. L. Larger, M.C. Soriano, D. Brunner, L. Appeltant, J.M. Gutiérrez, L. Pesquera, C.R. Mirasso, I. Fischer, Photonic information processing beyond turing: an optoelectronic implementation of reservoir computing. *Opt. Express* **20**(3), 3241–3249 (2012)
64. D. Brunner, M.C. Soriano, C.R. Mirasso, I. Fischer, Parallel photonic information processing at gigabyte per second data rates using transient states. *Nat. Commun.* **4**(1), 1364 (2013)
65. K. Vandoorne, P. Mechet, T. Van Vaerenbergh, M. Fiers, G. Morthier, D. Verstraeten, B. Schrauwen, J. Dambre, P. Bienstman, Experimental demonstration of reservoir computing on a silicon photonics chip. *Nat. Commun.* **5**(1), 3541 (2014)
66. Q. Vinckier, F. Duport, A. Smerieri, K. Vandoorne, P. Bienstman, M. Haelterman, S. Massar, High-performance photonic reservoir computer based on a coherently driven passive cavity. *Optica* **2**(5), 438–446 (2015)
67. L. Larger, A. Baylón-Fuentes, R. Martinenghi, V.S. Udaltsov, Y.K. Chembo, M. Jacquot, High-speed photonic reservoir computing using a time-delay-based architecture: million words per second classification. *Phys. Rev. X* **7**(1), 011015 (2017)
68. P. Antonik, N. Marsal, D. Brunner, D. Rontani, Human action recognition with a large-scale brain-inspired photonic computer. *Nat. Mach. Intell.* **1**(11), 530–537 (2019)
69. M. Rafayelyan, J. Dong, Y. Tan, F. Krzakala, S. Gigan, Large-scale optical reservoir computing for spatiotemporal chaotic systems prediction. *Phys. Rev. X* **10**(4), 041037 (2020)
70. S. Sunada, A. Uchida, Photonic neural field on a silicon chip: large-scale, high-speed neuro-inspired computing and sensing. *Optica* **8**(11), 1388–1396 (2021)
71. M. Nakajima, K. Tanaka, T. Hashimoto, Scalable reservoir computing on coherent linear photonic processor. *Commun. Phys.* **4**(1), 20 (2021)
72. A. Lupo, E. Picco, M. Zajnulina, S. Massar, Deep photonic reservoir computer based on frequency multiplexing with fully analog connection between layers. *Optica* **10**(11), 1478–1485 (2023)
73. Y.-W. Shen, R.-Q. Li, G.-T. Liu, J. Yu, X. He, L. Yi, C. Wang, Deep photonic reservoir computing recurrent network. *Optica* **10**(12), 1745–1751 (2023)
74. D. Wang, Y. Nie, G. Hu, H.K. Tsang, C. Huang, Ultrafast silicon photonic reservoir computing engine delivering over 200 tops. *Nat. Commun.* **15**(1), 10841 (2024)
75. Y. Nie, B. Yang, D. Wang, T. Wang, J. Wang, Z. Wang, C. Huang, Integrated laser graded neuron enabling high-speed reservoir computing without a feedback loop. *Optica* **11**(12), 1690–1699 (2024)

76. U. Teğin, M. Yıldırım, İ. Oğuz, C. Moser, D. Psaltis, Scalable optical learning operator. *Nat. Comput. Sci.* **1**(8), 542–549 (2021)
77. D. Pierangeli, G. Marucci, C. Conti, Photonic extreme learning machine by free-space optical propagation. *Photonics Res.* **9**(8), 1446–1454 (2021)
78. A. Lupo, L. Butschek, S. Massar, Photonic extreme learning machine based on frequency multiplexing. *Opt. Express* **29**(18), 28257–28276 (2021)
79. S. Saeed, M. Müftüoğlu, G.R. Cheeran, T. Bocklitz, B. Fischer, M. Chemnitz, Nonlinear inference capacity of fiber-optical extreme learning machines. *Nanophotonics* (2025). <https://doi.org/10.1515/nanoph-2025-0045>
80. S. Biasi, R. Franchi, L. Cerini, L. Pavesi, An array of microresonators as a photonic extreme learning machine. *APL Photonics* **8**(9), 096105 (2023)
81. H. Wang, J. Hu, A. Morandi, A. Nardi, F. Xia, X. Li, R. Savo, Q. Liu, R. Grange, S. Gigan, Large-scale photonic computing with nonlinear disordered media. *Nat. Comput. Sci.* **4**(6), 429–439 (2024)
82. A. Saade, F. Caltagirone, I. Carron, L. Daudet, A. Drémeau, S. Gigan, F. Krzakala, Random projections through multiple optical scattering: Approximating kernels at the speed of light. In: 2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), (IEEE, 2016), p. 6215–6219
83. I. Kang, S. Pang, Q. Zhang, N. Fang, G. Barbastathis, Recurrent neural network reveals transparent objects through scattering media. *Opt. Express* **29**(4), 5316–5326 (2021)
84. G.-B. Huang, Q.-Y. Zhu, C.-K. Siew, Extreme learning machine: theory and applications. *Neurocomputing* **70**(1–3), 489–501 (2006)
85. J. Wang, S. Lu, S.-H. Wang, Y.-D. Zhang, A review on extreme learning machine. *Multimed. Tools Appl.* **81**(29), 41611–41660 (2022)
86. S. Ding, H. Zhao, Y. Zhang, X. Xu, R. Nie, Extreme learning machine: algorithm, theory and applications. *Artif. Intell. Rev.* **44**(1), 103–115 (2015)
87. Y. Qing, Y. Zeng, Y. Li, G.-B. Huang, Deep and wide feature based extreme learning machine for image classification. *Neurocomputing* **412**, 426–436 (2020)
88. E. Soria-Olivas, J. Gomez-Sanchis, J.D. Martin, J. Vila-Frances, M. Martinez, J.R. Magdalena, A.J. Serrano, Belm: Bayesian extreme learning machine. *IEEE Trans. Neural Netw.* **22**(3), 505–509 (2011)
89. H. Zhou, G.-B. Huang, Z. Lin, H. Wang, Y.C. Soh, Stacked extreme learning machines. *IEEE Trans. Cybern.* **45**(9), 2013–2025 (2014)
90. J.R. Rausell-Campo, A. Hurtado, D. Pérez-López, J. Capmany Francoy, Programmable photonic extreme learning machines. *Laser & Photonics Reviews* **19**(9), 2400870 (2025)
91. A. Jacot, F. Gabriel, C. Hongler, Neural tangent kernel: Convergence and generalization in neural networks. *Advances in neural information processing systems*, vol. 31 (2018)
92. M. Gabrí, A. Manoel, C. Luneau, N. Macris, F. Krzakala, L. Zdeborová, et al. Entropy and mutual information in models of deep neural networks. *Advances in neural information processing systems*, vol. 31 (2018)
93. S. Yu, K. Wickstrøm, R. Jenssen, J.C. Principe, Understanding convolutional neural networks with information theory: an initial exploration. *IEEE Trans. Neural Netw. Learn. Syst.* **32**(1), 435–442 (2020)
94. D.-Y. Tsai, Y. Lee, E. Matsuyama, Information entropy measure for evaluation of image quality. *J. Digit. Imaging* **21**(3), 338–347 (2008)
95. M. Tancik, P. Srinivasan, B. Mildenhall, S. Fridovich-Keil, N. Raghavan, U. Singhal, R. Ramamoorthi, J. Barron, R. Ng, Fourier features let networks learn high frequency functions in low dimensional domains. *Adv. Neural. Inf. Process. Syst.* **33**, 7537–7547 (2020)
96. K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, (2016), p. 770–778
97. A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A.C. Berg, W.-Y. Lo, Segment anything. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, (2023), p. 4015–4026
98. T. Rahman, A. Khandakar, Y. Qiblawey, A. Tahir, S. Kiranyaz, S.B.A. Kashem, M.T. Islam, S. Al Maadeed, S.M. Zughaier, M.S. Khan, Exploring the effect of image enhancement techniques on COVID-19 detection using chest x-ray images. *Comput. Biol. Med.* **132**, 104319 (2021)
99. I. Oguz, J.-L. Hsieh, N.U. Dinc, U. Teğin, M. Yıldırım, C. Gigli, C. Moser, D. Psaltis, Programming nonlinear propagation for efficient optical learning machines. *Adv. Photonics* **6**(1), 016002–016002 (2024)
100. L. Maaten, G. Hinton, Visualizing data using t-sne. *J. Mach. Learn. Res.* **9**(11), 2579 (2008)
101. X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, R.M. Summers, Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, (2017), p. 2097–2106
102. A.E. Flanders, L.M. Prevedello, G. Shih, S.S. Halabi, J. Kalpathy-Cramer, R. Ball, J.T. Mongan, A. Stein, F.C. Kitamura, M.P. Lungren, Construction of a machine learning dataset through collaboration: the rsna 2019 brain ct hemorrhage challenge. *Radiology: Artificial Intelligence* **2**(3), 190211 (2020)
103. J. Bueno, S. Maktoobi, L. Froehly, I. Fischer, M. Jacquot, L. Larger, D. Brunner, Reinforcement learning in a large-scale photonic recurrent neural network. *Optica* **5**(6), 756–760 (2018)
104. C. Schudt, I. Laptev, B. Caputo, Recognizing human actions: a local svm approach. In: *Proceedings of the 17th International Conference on Pattern Recognition*, 2004. ICPR 2004, vol. 3, (IEEE, 2004), p. 32–36.
105. A. Gilbert, J. Illingworth, R. Bowden, Action recognition using mined hierarchical compound features. *IEEE Trans. Pattern Anal. Mach. Intell.* **33**(5), 883–897 (2010)
106. A. Grushin, D.D. Monner, J.A. Reggia, A. Mishra, Robust human action recognition via long short-term memory. In: *The 2013 International Joint Conference on Neural Networks (IJCNN)*, (IEEE, 2013), p. 1–8.
107. G. Campanella, M.G. Hanna, L. Geneslaw, A. Mirafior, V. Werneck Krauss Silva, K.J. Busam, E. Brogi, V.E. Reuter, D.S. Klimstra, T.J. Fuchs, Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nat. Med.* **25**(8), 1301–1309 (2019)
108. B.E. Bejnordi, M. Veta, P.J. Van Diest, B. Van Ginneken, N. Karssemeijer, G. Litjens, J.A. Van Der Laak, M. Hermesen, Q.F. Manson, M. Balkenhol, Diagnostic assessment of deep learning algorithms for detection of lymph node metastases in women with breast cancer. *JAMA* **318**(22), 2199–2210 (2017)
109. N. Otsu, A threshold selection method from gray-level histograms. *Automatica* **11**(285–296), 23–27 (1975)
110. R. Paniagua-Dominguez, Y.F. Yu, E. Khaidarov, S. Choi, V. Leong, R.M. Bakker, X. Liang, Y.H. Fu, V. Valuckas, L.A. Krivitsky, A metalens with a near-unity numerical aperture. *Nano Lett.* **18**(3), 2124–2132 (2018)
111. F. Zhou, Y. Chai, Near-sensor and in-sensor computing. *Nat. Electron.* **3**(11), 664–671 (2020)
112. Y. Chen, M. Nazhamaiti, H. Xu, Y. Meng, T. Zhou, G. Li, J. Fan, Q. Wei, J. Wu, F. Qiao, All-analog photoelectronic chip for high-speed vision tasks. *Nature* **623**(7985), 48–57 (2023)